IGCSE 0607 Practice Questions — E5.4 — Combined Questions & Markscheme

Symmetry

Total Questions: 21 — **Total Marks:** 131 Marks

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- **E5.4** Symmetry
  → **Paper 2 (Non-Calculator)** *(12 Qs)* · **Paper 4 (Calculator)** *(9 Qs)*

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1 Answer the following questions about the symmetry of two-dimensional shapes.

(a) Write down the number of lines of symmetry of a regular nonagon (a polygon with 9 sides).

..... **[1]**

(b) Write down the order of rotational symmetry of a parallelogram.

..... **[1]**

(c) Write down the mathematical name of a triangle that has exactly 1 line of symmetry and rotational symmetry of order 1.

..... **[1]**

2 Answer the following questions about 2D symmetry.

(a) Write down the order of rotational symmetry of a regular decagon.

..... **[1]**

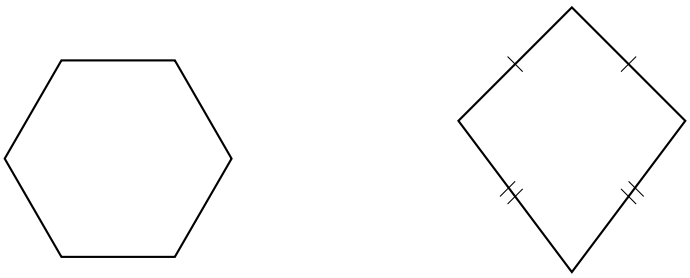
(b) A quadrilateral has exactly two lines of symmetry and rotational symmetry of order 2.
State the mathematical name of this quadrilateral.

..... **[1]**

(c) A shape has rotational symmetry of order 5.
Calculate the smallest angle of rotation that maps the shape onto itself.

..... ° **[1]**

3 The diagram shows a regular hexagon and a kite.



(a) For the regular hexagon, write down:

(i) the number of lines of symmetry,

..... **[1]**

(ii) the order of rotational symmetry.

..... **[1]**

(b) For the kite, write down:

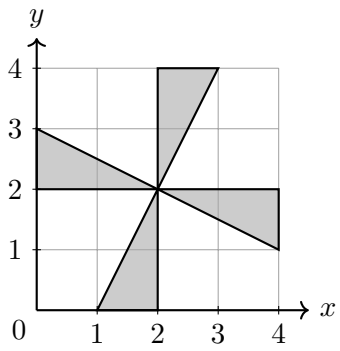
(i) the number of lines of symmetry,

..... **[1]**

(ii) the order of rotational symmetry.

..... **[1]**

4 The diagram shows a shaded shape on a coordinate grid.



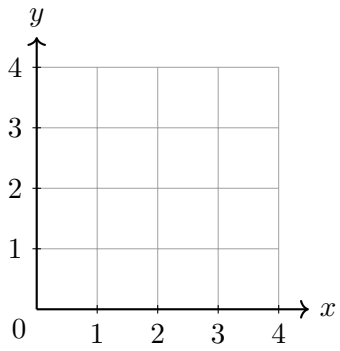
(a) Write down the order of rotational symmetry of the shaded shape.

..... **[1]**

(b) Write down the number of lines of symmetry of the shaded shape.

..... **[1]**

(c) On the grid below, draw the image of the shape after a reflection in the line $x = 2$.



[2]

5 Write down the symmetry properties for the following geometric shapes.

(a) A regular polygon has 15 sides.

Write down:

(i) the number of lines of symmetry,

..... **[1]**

(ii) the order of rotational symmetry.

..... **[1]**

(b) A quadrilateral has exactly two lines of symmetry and rotational symmetry of order 2.

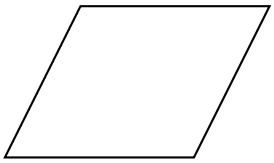
Write down the mathematical name of this quadrilateral.

..... **[1]**

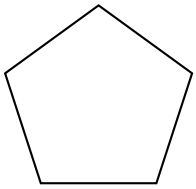
(c) Write down the number of planes of symmetry of a solid right pyramid with a regular pentagonal base.

..... **[1]**

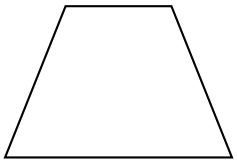
6 Three 2D shapes are shown below.



Parallelogram



Regular Pentagon

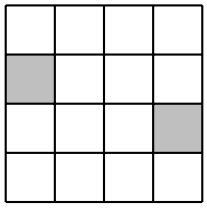


Isosceles Trapezium

Complete the table to show the number of lines of symmetry and the order of rotational symmetry for each shape.

[6]

7 The diagram shows a 4×4 grid with two squares shaded.



(a) For the diagram above, write down

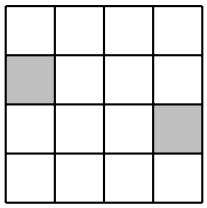
(i) the number of lines of symmetry,

..... **[1]**

(ii) the order of rotational symmetry.

..... **[1]**

(b) On the grid below, shade two more squares so that the completed pattern has exactly 2 lines of symmetry.

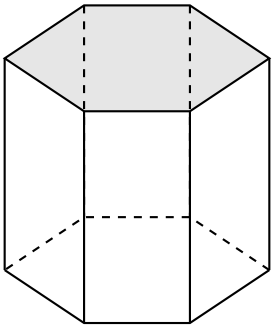


[2]

(c) A regular polygon has rotational symmetry of order 12.
Write down the number of lines of symmetry of this regular polygon and calculate the size of each exterior angle.

Lines of symmetry = ... , Exterior angle = ... **[2]**

8 The diagram shows a solid regular hexagonal prism.



(a) Write down the number of planes of symmetry of this regular hexagonal prism.

..... **[2]**

(b) Write down the order of rotational symmetry of the prism about the vertical axis passing through the centres of its hexagonal faces.

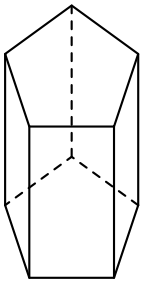
..... **[1]**

(c) A different solid is a right square-based pyramid.

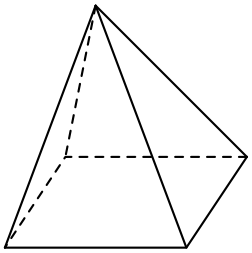
Write down the number of planes of symmetry of this pyramid.

..... **[1]**

9 Solid A is a right regular pentagonal prism.
Solid B is a right square-based pyramid.



Solid A



Solid B

(a) State the number of planes of symmetry for:

(i) Solid A,

..... **[1]**

(ii) Solid B,

..... **[1]**

(iii) a solid cylinder.

..... **[1]**

(b) A regular polygon has exactly 12 lines of symmetry.
Calculate the size of one of its interior angles.

..... ° **[2]**

10 Consider the symmetry properties of the following 3D solid shapes.

(a) A right square-based pyramid has a base of side length 8 cm and perpendicular height 10 cm.
Write down the number of planes of symmetry for this pyramid.

..... **[1]**

(b) Write down the total number of planes of symmetry of a solid regular pentagonal prism.

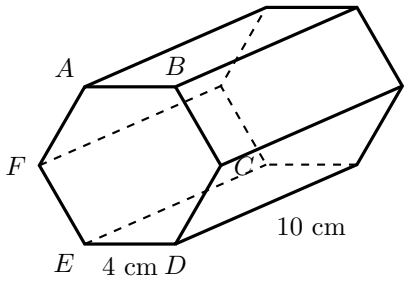
..... **[2]**

(c) A right circular cone has base radius r and slant height l.
Describe fully its axis of symmetry.

..... **[2]**

11 The diagram shows a right prism whose cross-section is a regular hexagon $ABCDEF$ of side length 4 cm. The length of the prism is 10 cm.

NOT TO SCALE



(a) For this regular hexagonal prism, write down

(i) the total number of planes of symmetry,

..... **[2]**

(ii) the order of rotational symmetry about the axis passing through the centres of the two hexagonal faces.

..... **[1]**

(b) A solid right pyramid has a regular pentagon as its base.
Write down the number of planes of symmetry and the order of rotational symmetry of this pyramid.

Planes of symmetry = ... , Order of rotational symmetry = ...
[2]

(c) A solid right circular cone has base radius 6 cm and vertical height 8 cm. State the number of planes of symmetry of the cone, and calculate the total surface area of the cone, in cm^2, in terms of π.

Planes of symmetry = ... , Total surface area = ... **[3]**

12 The points $A(1, 6)$ and $B(7, 4)$ are two vertices of a quadrilateral $ABCD$.
The point $M(3, 2)$ is the centre of rotational symmetry of $ABCD$.
The quadrilateral $ABCD$ has rotational symmetry of order 4.

(a) Find the coordinates of C.

(..... , ...) **[2]**

(b) The line AC is a line of symmetry of the quadrilateral.

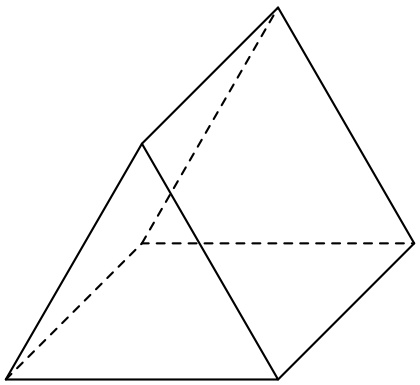
Find the equation of the line AC.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

(c) Calculate the exact area of the quadrilateral $ABCD$.

..... **[3]**

13 The diagram shows a solid right prism.
The cross-section of the prism is an equilateral triangle.



(a) State the number of planes of symmetry of this solid prism.

..... **[2]**

(b) State the order of rotational symmetry of the prism about the axis passing through the centroids of the two equilateral triangular faces.

..... **[1]**

(c) A different solid is a right regular hexagonal prism.

State the number of planes of symmetry of this hexagonal prism.

..... **[2]**

14 **(a)** A regular polygon has an exterior angle of 15°.

(i) Calculate the number of sides of this polygon.

..... **[2]**

(ii) Write down the number of lines of symmetry of this polygon.

..... **[1]**

(b) Write down the mathematical name of a quadrilateral that has exactly 1 line of symmetry and rotational symmetry of order 1.

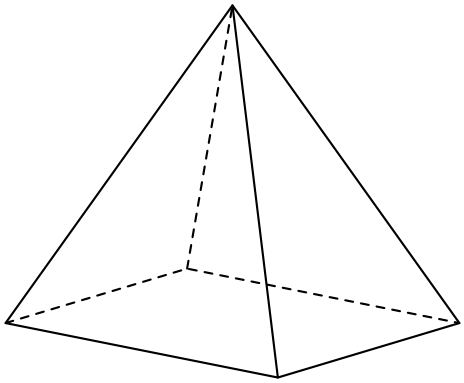
..... **[1]**

(c) A solid cone has a base radius of 5 cm and a perpendicular height of 12 cm.

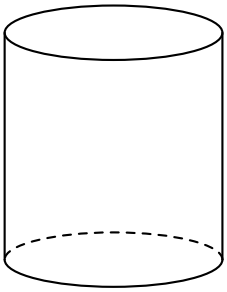
State the number of planes of symmetry of this cone.

..... **[1]**

15 The diagram shows a solid square-based pyramid and a solid cylinder.



Square-based pyramid



Cylinder

(a) State the number of planes of symmetry for each solid.

(i) The solid square-based pyramid.

..... **[1]**

(ii) The solid cylinder.

..... **[1]**

(b) A solid square-based pyramid has a base side length of 6 cm.
The pyramid has exactly 4 planes of symmetry and its volume is 48 cm^3.
Calculate the length of a sloping edge of this pyramid.

..... cm **[5]**

16 A rhombus $ABCD$ has its centre at the origin $(0, 0)$.

(a) The line of symmetry passing through vertices A and C has the equation $y = 3x$.
Find the equation of the line of symmetry passing through vertices B and D.

$y =$ ..... **[2]**

(b) Write down the order of rotational symmetry of the rhombus $ABCD$.

..... **[1]**

(c) A different shape is a regular polygon with n sides.
The number of lines of symmetry of this polygon is equal to n.
The interior angle of this polygon is $165°$.

Calculate the value of n.

$n =$ ..... **[3]**

17 **(a)** A solid wooden block is in the shape of a right prism.
The cross-section of the prism is a regular octagon.

(i) Write down the order of rotational symmetry of the cross-section.

..... **[1]**

(ii) State the total number of planes of symmetry of this right prism.

..... **[2]**

(b) A different solid is a right pyramid with a regular polygon of n sides as its base.
The pyramid has exactly 12 planes of symmetry.

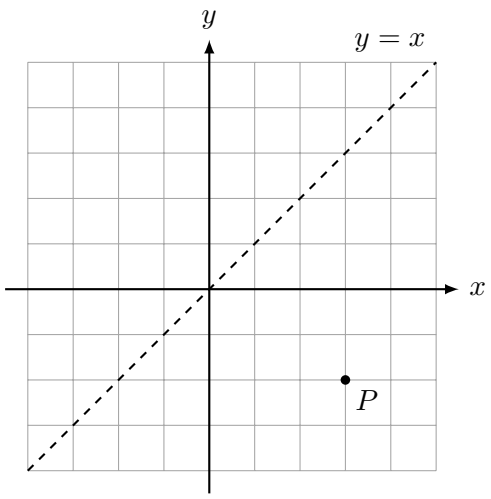
(i) Write down the value of n.

$n =$ ..... **[1]**

(ii) Calculate the sum of the interior angles of the base of this pyramid.

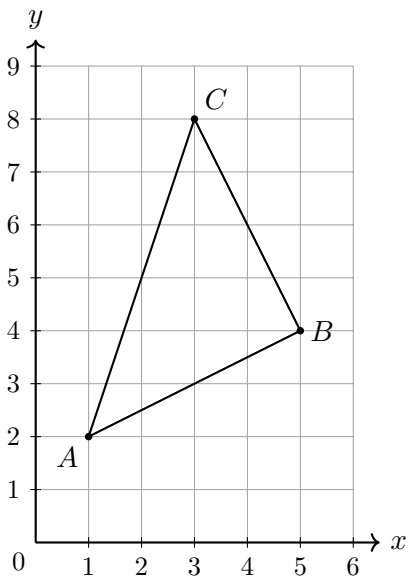
..... ° **[2]**

(c) A shape S is drawn on a coordinate grid.
The shape S has a line of symmetry with equation $y = x$.
One vertex of the shape is at the point $P(3, -2)$.
Write down the coordinates of the image of P after a reflection in this line of symmetry.



(..... , ...) **[2]**

18 Triangle ABC has vertices at $A(1, 2)$, $B(5, 4)$ and $C(3, 8)$.



(a) Show that triangle ABC is an isosceles triangle.

[3]

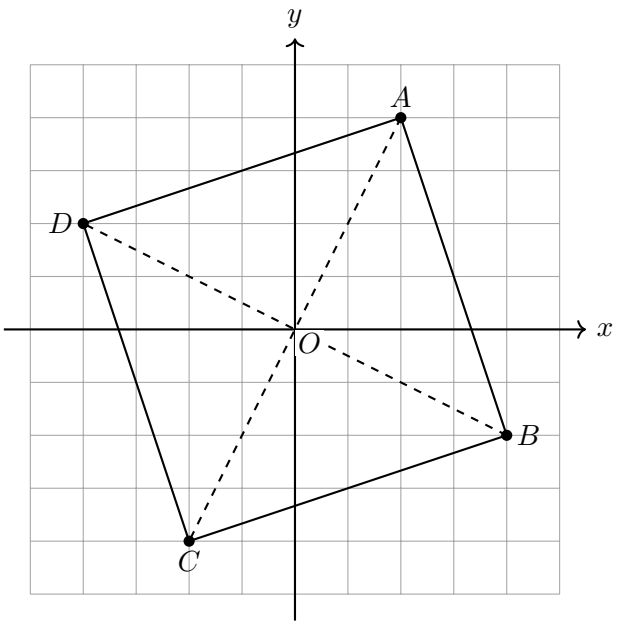
(b) Triangle ABC has exactly one line of symmetry.
Find the equation of this line of symmetry.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[4]**

(c) A different shape is a regular polygon with n sides.
It has exactly 15 lines of symmetry.
Calculate the size of one interior angle of this polygon.

..... ° **[2]**

19 The diagram shows a rhombus $ABCD$.
The diagonals AC and BD intersect at the origin $O(0, 0)$.
The line of symmetry passing through A and C has the equation $y = 2x$.



(a) Write down the order of rotational symmetry of the rhombus $ABCD$.

..... **[1]**

(b) Find the equation of the line of symmetry passing through B and D.

$y =$ ..... **[2]**

(c) The coordinates of A are $(k, 2k)$, where $k > 0$.
The area of the rhombus is 40 square units, and the length of the diagonal BD is $4\sqrt{5}$ units.

(i) Show that the length of the diagonal AC is $4\sqrt{5}$ units.

[2]

(ii) Calculate the coordinates of A.

$A = (\ldots\ldots\ ,\ \ldots\ldots)$ **[2]**

(iii) Calculate the coordinates of B.

$B = (\ldots\ldots\ ,\ \ldots\ldots)$ **[3]**

20 A right solid prism has a uniform cross-section which is a regular polygon with n sides.
The prism has exactly 7 planes of symmetry.
The length of the prism is not equal to its width.

(a) Explain why $n = 6$.

[2]

(b) The longest diagonal of the hexagonal cross-section is 12 cm.
The volume of the prism is 450 cm^3.

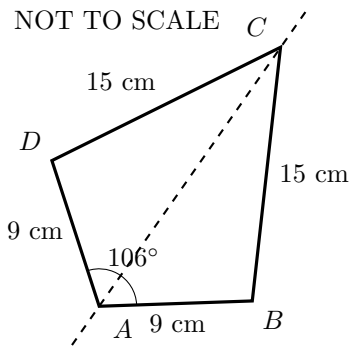
(i) Calculate the length of the prism.

..... cm **[4]**

(ii) Calculate the total surface area of the prism.

..... cm^2 **[4]**

21 The diagram shows a quadrilateral $ABCD$ that has exactly one line of symmetry, which passes through vertices A and C.
$AB = AD = 9$ cm, $BC = CD = 15$ cm, and angle $BAD = 106°$.



(a) Write down the special mathematical name for quadrilateral $ABCD$ and state its order of rotational symmetry.

Name: ..... , Order of rotational symmetry = ... **[2]**

(b) The diagonals AC and BD intersect at M.
Using the symmetry of $ABCD$, explain why angle $AMB = 90°$.

..... **[1]**

(c) Calculate

(i) the length of diagonal BD, giving your answer correct to 3 significant figures,

$BD =$ ..... cm **[3]**

(ii) the size of angle BCD, giving your answer correct to 1 decimal place.

Angle $BCD =$ ..... ° **[3]**

(d) A right prism has quadrilateral $ABCD$ as its uniform cross-section.
Write down the total number of planes of symmetry of this prism.

..... **[2]**

1 **(a)** 9 **[B1]**

(b) 2 **[B1]**

(c) Isosceles triangle **[B1]**

[Back to Question] *[Total Marks: 3]*

2 **(a)** 10 **[B1]**

(b) Rhombus OR Rectangle **[B1]**

(c) 72 **[B1]**

[Back to Question] *[Total Marks: 3]*

3 **(a)** **(i)** 6 **[B1]**

(ii) 6 **[B1]**

(b) **(i)** 1 **[B1]**

(ii) 1 **[B1]**

[Back to Question] *[Total Marks: 4]*

4 **(a)** 4 **[B1]**

(b) 0 **[B1]**

(c) Correct reflection of at least two 'petals' (e.g. vertices at $(1, 4)$ and $(4, 3)$) **[M1]**

Fully correct shape drawn **[A1]**

[Back to Question] *[Total Marks: 4]*

5 **(a)** **(i)** 15 **[B1]**

(ii) 15 **[B1]**

(b) Rectangle OR Rhombus **[B1]**

(c) 5 **[B1]**

[Back to Question] *[Total Marks: 4]*

6 Parallelogram: 0 lines, order 2 **[B2, B1 for each correct value]**

Regular Pentagon: 5 lines, order 5 **[B2, B1 for each correct value]**

Isosceles Trapezium: 1 line, order 1 **[B2, B1 for each correct value]**

[Back to Question] *[Total Marks: 6]*

7 **(a)** **(i)** 0 **[B1]**

(ii) 2 **[B1]**

(b) Both squares in the left column row 2 and right column row 3 shaded (or top row column 2 and bottom row column 3 shaded) **[B2]**

(c) Lines of symmetry = 12 **[B1]**

Exterior angle = 30° **[B1]**

[Back to Question] *[Total Marks: 6]*

8 **(a)** Identifies 6 vertical planes of symmetry or 1 horizontal plane of symmetry **[M1]**

7 **[A1]**

(b) 6 **[B1]**

(c) 4 **[B1]**

[Back to Question] *[Total Marks: 4]*

9 **(a)** **(i)** 6 **[B1]**

(ii) 4 **[B1]**

(iii) Infinite **[B1]**

(b) Exterior angle $= \frac{360}{12} = 30°$ **[M1]**

Interior angle $= 180 - 30 = 150$ **[A1]**

[Back to Question] *[Total Marks: 5]*

10 **(a)** 4 **[B1]**

(b) 6 **[B2]**

Strictly 5 (vertical) or 1 (horizontal) planes identified **[B1]**

(c) A straight line passing through the apex (top vertex) **[B1]**

and the centre of the circular base. **[B1]**

[Back to Question] *[Total Marks: 5]*

11 **(a)** **(i)** 6 longitudinal planes and 1 perpendicular bisecting plane seen or implied **[M1]**

7 **[A1]**

(ii) 6 **[B1]**

(b) Planes of symmetry = 5 **[B1]**

Order of rotational symmetry = 5 **[B1]**

(c) Planes of symmetry = infinite (or ∞) **[B1]**

Slant height $l = \sqrt{6^2 + 8^2} = 10$ cm **[M1]**

Total surface area $= \pi(6)(10) + \pi(6)^2 = 96\pi$ **[A1]**

[Back to Question] *[Total Marks: 8]*

12 **(a)** Recognises that order 4 symmetry implies point symmetry (order 2) about M, so M is the midpoint of AC **[M1]**

$(5, -2)$ **[A1]**

(b) Gradient of $AC = \dfrac{-2-6}{5-1}$ or $\dfrac{-8}{4} = -2$ **[M1]**

Substitute $(1, 6)$ or $(5, -2)$ or $(3, 2)$ into $y = -2x + c$ OR $y - 6 = -2(x - 1)$ **[M1]**

$-2x + 8$ **[A1]**

(c) Deduces that $ABCD$ must be a square because it has order 4 rotational symmetry (area is $|AB|^2$) **[M1]**

$|AB|^2 = (7-1)^2 + (4-6)^2$ **[M1]**

40 **[A1]**

[Back to Question] *[Total Marks: 8]*

13 **(a)** 4 **[B2]**

If 0 scored, allow [B1] for identifying either 3 (vertical) OR 1 (horizontal/mid-plane).

(b) 3 **[B1]**

(c) 7 **[B2]**

If 0 scored, allow [B1] for 6.

[Back to Question] *[Total Marks: 5]*

14 **(a)** **(i)** $\dfrac{360}{15}$ **[M1]**

24 **[A1]**

(ii) 24 **[B1]**

(b) Kite OR Isosceles Trapezium (accept Isosceles Trapezoid) **[B1]**

(c) Infinite (accept an infinite number) **[B1]**

[Back to Question] *[Total Marks: 5]*

15 **(a)** **(i)** 4 **[B1]**

(ii) Infinite (or infinitely many) **[B1]**

(b) Uses volume formula: $\dfrac{1}{3} \times 6^2 \times h = 48$ **[M1]**

Solves for height: $h = 4$ **[A1]**

Calculates distance from centre of base to a corner $= \sqrt{3^2+3^2} = \sqrt{18}$ (or $3\sqrt{2}$ or 4.24...) **[M1]**

Uses Pythagoras' theorem for sloping edge $l = \sqrt{4^2 + (\sqrt{18})^2}$ **[M1]**

5.83 (or 5.830...) **[A1]**

[Back to Question] *[Total Marks: 7]*

16 **(a)** Understanding that diagonals of a rhombus intersect at 90°, hence gradient $m = -\frac{1}{3}$ **[M1]**

$-\frac{1}{3}x$ (accept equivalent forms) **[A1]**

(b) 2 **[B1]**

(c) Exterior angle $= 180° - 165° = 15°$ **[M1]**

$n = \frac{360}{15}$ **[M1]**

$n = 24$ **[A1]**

[Back to Question] *[Total Marks: 6]*

17 **(a)** **(i)** 8 **[B1]**

(ii) 9 **[B2]**

If B0, score B1 for 8 (identifying only the vertical planes)

(b) **(i)** 12 **[B1]**

(ii) $(12-2) \times 180$ **[M1]**

1800 **[A1]**

(c) $(-2, 3)$ **[B2]**

If B0, score M1 for point plotted correctly on grid OR one coordinate correct

[Back to Question] *[Total Marks: 8]*

18 **(a)** Calculates length $AB = \sqrt{(5-1)^2+(4-2)^2} = \sqrt{20}$ **[M1]**

Calculates length $BC = \sqrt{(3-5)^2+(8-4)^2} = \sqrt{20}$ **[M1]**

States $AB = BC$ therefore the triangle is isosceles **[A1]**

(b) Identifies midpoint of $AC = (\frac{1+3}{2}, \frac{2+8}{2}) = (2, 5)$ **[M1]**

Calculates gradient of line of symmetry $= \frac{4-5}{5-2} = -\frac{1}{3}$ (or uses perpendicular to AC) **[M1]**

Substitutes $(5, 4)$ or $(2, 5)$ into $y = -\frac{1}{3}x + c$ to find c **[M1]**

$-\frac{1}{3}x + \frac{17}{3}$ **[A1]**

(c) Identifies $n = 15$ and uses $\frac{(15-2) \times 180}{15}$ OR $180 - (360 \div 15)$ **[M1]**

156 **[A1]**

[Back to Question] *[Total Marks: 9]*

19 **(a)** 2 **[B1]**

(b) Gradient of $BD = -\frac{1}{2}$ (using $m_1 \times m_2 = -1$) **[M1]**

$-0.5x$ or $-\frac{1}{2}x$ **[A1]**

(c) **(i)** Uses Area $= \frac{1}{2} \times d_1 \times d_2$: $\frac{1}{2} \times AC \times 4\sqrt{5} = 40$ **[M1]**

$AC = \frac{80}{4\sqrt{5}} = 4\sqrt{5}$ **[A1]**

(ii) $OA = 2\sqrt{5}$, so $k^2 + (2k)^2 = (2\sqrt{5})^2$ or $5k^2 = 20$ **[M1]**

$(2, 4)$ **[A1]**

(iii) Sets up distance equation for $B(x, -0.5x)$ with $OB = 2\sqrt{5}$: $x^2 + (-0.5x)^2 = 20$ **[M1]**

$1.25x^2 = 20$ giving $x^2 = 16$ **[M1]**

$(4, -2)$ (rejects negative x due to visual quadrant) **[A1]**

[Back to Question] *[Total Marks: 10]*

20 **(a)** States that a right prism with a regular n-gon base has n vertical planes of symmetry and 1 horizontal plane of symmetry (halfway along its length) **[B1]**

Sets up the equation $n + 1 = 7$, yielding $n = 6$ **[B1]**

(b) **(i)** Recognises the side length of the hexagon $= 12 \div 2 = 6$ cm **[B1]**

Calculates area of the hexagonal cross-section $= 6 \times \left(\frac{1}{2} \times 6 \times 6 \times \sin(60°)\right) = 54\sqrt{3}$ (or 93.53...) **[M1]**

Uses Volume = Area × Length to find Length $= \frac{450}{54\sqrt{3}}$ **[M1]**

4.81 (or 4.811...) **[A1]**

(ii) Method for total area = 2 × Base Area + 6 × Lateral Rectangular Area **[M1]**

Substitutes values: $2 \times 54\sqrt{3} + 6 \times (6 \times 4.811...)$ **[M1]**

Simplifies to $108\sqrt{3} + 100\sqrt{3}$ (or 187.06... + 173.2...) **[M1]**

360 (or 360.2... to 360.3...) **[A1]**

[Back to Question] *[Total Marks: 10]*

21 **(a)** Name: kite **[B1]**

Order of rotational symmetry = 1 **[B1]**

(b) The line of symmetry AC is the perpendicular bisector of BD (or B is the reflection of D in AC) **[B1]**

(c) **(i)** Angle $BAM = \dfrac{106^\circ}{2} = 53^\circ$ **[B1]**

$BM = 9\sin 53^\circ \ (= 7.1877\ldots)$ **[M1]**

$BD = 2 \times 7.1877\cdots = 14.4$ **[A1]**

(ii) $\sin(\text{angle } BCM) = \dfrac{7.1877\ldots}{15} \ (= 0.47918\ldots)$ **[M1]**

Angle $BCM = 28.63^\circ$ **[A1]**

Angle $BCD = 2 \times 28.63^\circ = 57.3$ **[A1]**

(d) One plane along AC through the length of the prism and one plane bisecting the prism perpendicular to its length **[M1]**

2 **[A1]**

[Back to Question] *[Total Marks: 11]*